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Required (Core) Practicals — CP1 to CP8

Edexcel GCSE (9-1) Chemistry (1CH0) — Assessed in the written exams

Edexcel GCSE Chemistry specifies eight mandatory Core Practicals (CPs). The specification labels each one by its specification point (shown in brackets below); CP1–CP8 simply follows the specification's order. Points ending in "C" (5.9C, 9.6C, 9.28C) are in the separate Chemistry course only. You will not be assessed on performing them in an exam, but you **must** know the method, variables, measurements, analysis, sources of error and improvements for each. They belong to Topics 2, 3, 5, 7 and 9.

CP1 (2.11) — Investigating the composition of inks using simple distillation and paper chromatography

Apparatus: Ink sample, round-bottomed or distillation flask, anti-bumping granules, Bunsen burner (or heating mantle), tripod and gauze, thermometer, Liebig condenser with cold-water supply, beaker or test tube to collect the distillate, ice bath; chromatography paper, pencil, ruler, capillary tube, beaker, water (solvent), watch glass or lid.

Part 1 — Simple distillation (separating the solvent)

1. Pour about 10 cm^3 of ink into the flask and add a few anti-bumping granules.
2. Fit the thermometer so its bulb is level with the side arm (it measures the temperature of the vapour leaving the flask). Connect the condenser: cold water in at the bottom, out at the top.
3. Heat gently until the ink boils. The solvent evaporates, the vapour passes into the condenser, cools and condenses, and drips into the collecting tube.
4. Record the temperature while the distillate is collecting. A steady reading of about 100 °C shows the solvent is water.

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5. Stop heating before the flask boils dry. The dyes remain in the flask.

Part 2 — Paper chromatography (separating the dyes)

1. Draw a **pencil** start line about 2 cm from the bottom of the chromatography paper.
2. Use a capillary tube to put a small, concentrated spot of the ink on the line (and spots of known dyes for comparison).
3. Pour water into a beaker to a depth **below** the start line. Hang the paper so its bottom edge dips into the solvent; cover the beaker.
4. Leave until the solvent front is near the top. Remove the paper, mark the solvent front in pencil straight away and let it dry.
5. Measure the distance from the start line to the centre of each spot and to the solvent front.

$$R_f = \text{distance moved by substance} \div \text{distance moved by solvent}$$

Analysis

- The number of spots shows the minimum number of dyes in the ink. A pure substance gives one spot.
- The stationary phase is the paper; the mobile phase is the solvent (water). Dyes that are more soluble in the solvent, or attracted less to the paper, travel further and have higher R_f values.
- A dye can be identified by matching its R_f value (in the same solvent) with a known substance, or by spots at the same height as a reference.
- R_f values are always less than 1 and have no units.

⚠ **Safety:** Use anti-bumping granules and never heat a sealed apparatus. Heat gently, and do not let the flask boil dry. Inks may stain and some solvents are flammable — keep them away from flames.

Exam Tip: Explain the details: pencil for the start line because ink would dissolve and run; solvent below the start line so the spots do not wash off into the solvent; lid on to stop the solvent evaporating. A spot that stays on the start line is insoluble in that solvent.

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CP2 (3.6) — Investigating the change in pH on adding powdered calcium hydroxide or calcium oxide to dilute hydrochloric acid

Apparatus: Dilute hydrochloric acid (e.g. 0.05 mol/dm³), calcium hydroxide (or calcium oxide) powder, 100 cm³ beaker, 50 cm³ measuring cylinder, spatula, balance (or pre-weighed portions of 0.3 g), glass rod, universal indicator paper and colour chart (or a calibrated pH meter).

Method

1. Measure 50 cm³ of dilute hydrochloric acid into the beaker.
2. Measure the starting pH: dip the glass rod in the acid, touch it onto a piece of universal indicator paper and compare with the colour chart (or read a pH meter).
3. Add a measured small mass of calcium hydroxide powder (e.g. 0.3 g). Stir thoroughly.
4. Measure and record the pH again.
5. Repeat, adding the same mass each time, until the pH stops changing (around 8 portions).
6. Plot pH (y-axis) against total mass of calcium hydroxide added (x-axis).

$Ca(OH)_2(s) + 2HCl(aq) \rightarrow CaCl_2(aq) + 2H_2O(l)$ **(HT)** ionic: $H^+(aq) + OH^-(aq) \rightarrow H_2O(l)$

Analysis

- The pH starts low (about 1–2). It rises slowly at first, then rapidly near neutralisation (pH 7), then levels off at a value above 7 once excess base has been added.
- The final pH is only slightly alkaline (around 11–12 at most) because calcium hydroxide is only slightly soluble; with calcium oxide the curve has the same shape.
- The mass needed to reach pH 7 can be read from the graph and used to find the amount of base that neutralises the acid.

⚠ **Safety:** Wear eye protection — calcium hydroxide and calcium oxide are irritants (calcium oxide reacts exothermically with water). Avoid raising dust from the powders.

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Exam Tip: A pH meter gives a more precise and less subjective reading than universal indicator, whose colours must be judged by eye. Control variables: volume and concentration of acid, mass of each portion, stirring time before reading the pH.

CP3 (3.17) — Preparing pure, dry hydrated copper sulfate crystals from copper oxide

Apparatus: Dilute sulfuric acid, copper(II) oxide powder, 100 cm^3 beaker, measuring cylinder, spatula, glass rod, water bath (or beaker of hot water on a tripod and gauze, Bunsen burner), filter funnel and filter paper, conical flask, evaporating basin, filter paper or warm oven for drying.

Method

1. Measure 20 cm^3 of dilute sulfuric acid into a beaker and warm it gently in a water bath (heating speeds up the reaction).
2. Add copper(II) oxide a spatula at a time, stirring, until no more dissolves and black solid remains. The copper oxide is now **in excess**, so all the acid has reacted.
3. Filter the mixture to remove the excess copper oxide. The blue filtrate is copper sulfate solution.
4. Pour the filtrate into an evaporating basin and heat it over a water bath to evaporate some of the water until crystals start to form at the edge (the point of crystallisation).
5. Leave the solution in a cool place for the crystals to grow slowly.
6. Remove the crystals, pat them dry between filter papers (or dry in a warm oven) to get pure, dry hydrated crystals.

$$CuO(s) + H_2SO_4(aq) \rightarrow CuSO_4(aq) + H_2O(l)$$

Analysis

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- Using excess of the insoluble base makes sure no acid remains in the product; filtration then removes the unreacted base.
- The product is hydrated copper sulfate, $CuSO_4 \cdot 5H_2O$ — blue crystals that contain water of crystallisation.
- Percentage yield can be calculated: actual mass of crystals ÷ theoretical mass × 100. It is below 100% because some solution is left on the filter paper and in the basin, and some copper sulfate stays dissolved.

⚠ **Safety:** Wear eye protection. Copper sulfate and copper oxide are harmful; wash hands after use. Use a water bath rather than heating to dryness — strong heating would drive off the water of crystallisation and could cause spitting.

Exam Tip: The water bath is important: heating gently stops the solution spitting and stops the hydrated crystals turning into white anhydrous copper sulfate. Slow cooling gives larger crystals.

CP4 (3.31) — Investigating the electrolysis of copper sulfate solution with inert electrodes and copper electrodes

Apparatus: Copper sulfate solution, 100 cm^3 beaker, two graphite (carbon) electrodes, two copper electrodes, electrode holder, low-voltage d.c. power supply, ammeter, variable resistor, connecting leads and crocodile clips, balance, propanone (to rinse and dry copper electrodes), test tube and splint.

Part 1 — Inert (graphite) electrodes

1. Half-fill the beaker with copper sulfate solution and place the two graphite electrodes in it, not touching.
2. Connect them to the d.c. supply and switch on (e.g. 4 V).
3. Observe the electrodes. Collect any gas at the anode in an inverted test tube and test it with a glowing splint.

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- **Cathode (–):** a pink-brown coating of copper forms. **(HT)** $Cu^{2+}(aq) + 2e^- \rightarrow Cu(s)$
- **Anode (+):** bubbles of oxygen (relights a glowing splint). **(HT)** $4OH^-(aq) \rightarrow O_2(g) + 2H_2O(l) + 4e^-$
- The blue colour of the solution fades as Cu^{2+} ions are removed.

Part 2 — Copper electrodes

1. Clean two copper electrodes with emery paper, rinse with propanone, dry and weigh each one.
2. Set up the circuit with an ammeter and variable resistor in series. Pass a constant current (e.g. 0.2 A) for a fixed time (e.g. 20 minutes).
3. Remove the electrodes, rinse carefully with distilled water then propanone, allow to dry and reweigh.
4. Repeat with different currents (e.g. 0.3, 0.4, 0.5 A) using fresh or recleaned electrodes.

- **Cathode:** gains mass. **(HT)** $Cu^{2+}(aq) + 2e^- \rightarrow Cu(s)$
- **Anode:** loses mass. **(HT)** $Cu(s) \rightarrow Cu^{2+}(aq) + 2e^-$
- The blue colour stays the same because copper ions are added at the anode as fast as they are removed at the cathode (this is how copper is purified).

Analysis

- Plot change in mass of each electrode against current. Both give straight lines through the origin: mass change is directly proportional to current (for a fixed time).
- The mass gained by the cathode should equal the mass lost by the anode; in practice the anode often loses slightly more because impurities or loose copper fall off.

> ⚠ **Safety:** Wear eye protection. Use low voltage d.c. only. Propanone is highly flammable — no flames nearby. Copper sulfate is harmful; wash hands after use.

> **Exam Tip:** Errors: copper deposit flaking off when the cathode is rinsed, or not drying the electrodes fully before weighing (water adds mass). Keep the time, electrode distance, solution concentration and volume constant.

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CP5 (5.9C) — Carrying out an accurate acid–alkali titration

Apparatus: Burette (50 cm³) with clamp and stand, 25.0 cm³ volumetric pipette and pipette filler, conical flask, white tile, small funnel, phenolphthalein or methyl orange indicator, sodium hydroxide solution, dilute hydrochloric (or sulfuric) acid.

Method

1. Rinse the pipette with a little of the alkali, then use the pipette filler to transfer exactly 25.0 cm³ of sodium hydroxide solution to a clean conical flask.
2. Add a few drops of indicator (phenolphthalein: pink in alkali; methyl orange: yellow in alkali) and stand the flask on a white tile.
3. Rinse the burette with the acid, then fill it using a funnel. Remove the funnel and run a little acid through so the jet is full. Record the starting volume, reading from the bottom of the meniscus at eye level.
4. Add acid from the burette while swirling the flask. Do a rough (trial) titration first to find the approximate end-point.
5. Repeat accurately: add acid quickly to within about 2 cm³ of the rough value, then drop by drop until the indicator just changes colour permanently (phenolphthalein pink → colourless; methyl orange yellow → red/orange).
6. Record the final burette reading to the nearest 0.05 cm³. Titre = final – initial reading.
7. Repeat until you have **concordant** titres (within 0.10 cm³ of each other) and calculate the mean of these.

$NaOH(aq) + HCl(aq) \rightarrow NaCl(aq) + H_2O(l)$ **(HT)** moles = concentration (mol/dm³) × volume (dm³)

Analysis

(HT) Titration calculations (spec 5.8C, 5.10C) are Higher tier only:

- Calculate moles of the solution of known concentration from the mean titre (convert cm³ to dm³ by dividing by 1000).

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- Use the mole ratio in the balanced equation to find the moles of the other reactant (1:1 for HCl and NaOH; 1:2 for H_2SO_4 and NaOH).
- Concentration of the unknown = moles ÷ volume in dm^3. Concentration in g/dm^3 = mol/dm^3 × M_r.

Titration can also be used to make a pure salt: find the exact volumes needed, repeat without indicator, then crystallise.

⚠ **Safety:** Wear eye protection — sodium hydroxide is corrosive, even when dilute it is an irritant to eyes. Always use a pipette filler, never your mouth. Fill the burette below eye level.

Exam Tip: Do not use universal indicator — it changes colour gradually so there is no sharp end-point. Swirl constantly and add drop by drop near the end-point. Ignore the rough titre when calculating the mean.

CP6 (7.1) — Investigating the effect of changing conditions on the rate of reaction

The specification asks for two methods: (a) measuring the production of a gas, in the reaction between hydrochloric acid and marble chips, and (b) observing a colour change, in the reaction between sodium thiosulfate and hydrochloric acid.

Method (a) — Gas production: marble chips and hydrochloric acid

Apparatus: Conical flask, bung with delivery tube, gas syringe (or upturned measuring cylinder in a trough of water), stopwatch, balance, marble chips of different sizes, dilute hydrochloric acid, measuring cylinder.

1. Measure 50 cm^3 of dilute hydrochloric acid into the conical flask.
2. Add a measured mass of marble chips (e.g. 5 g of small chips), insert the bung immediately and start the stopwatch.
3. Record the volume of carbon dioxide collected every 30 s for about 5 minutes.

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4. Repeat with the same mass of medium and large chips (or with powdered calcium carbonate) to change surface area; or vary the acid concentration or temperature.
5. Plot volume of gas (y-axis) against time (x-axis) for each condition.

$$CaCO_3(s) + 2HCl(aq) \rightarrow CaCl_2(aq) + H_2O(l) + CO_2(g)$$

Alternative: put the flask on a balance with a cotton-wool plug and record the **loss in mass** as CO_2 escapes.

Method (b) — Colour change: sodium thiosulfate and hydrochloric acid

> **Apparatus:** Conical flask, paper marked with a black cross, measuring cylinders, stopwatch, thermometer, water bath (to change temperature), sodium thiosulfate solution, dilute hydrochloric acid.

1. Place the conical flask on the paper marked with a black cross.
2. Measure 50 cm^3 of sodium thiosulfate solution into the flask and record its temperature.
3. Add 10 cm^3 of dilute hydrochloric acid, swirl and start the stopwatch.
4. Look down through the solution and stop the clock when the cross can no longer be seen (the solution turns cloudy as a yellow precipitate of sulfur forms).
5. Repeat at different temperatures (warming the solutions in a water bath first) or with different concentrations of thiosulfate (diluting with water but keeping total volume the same).

$$Na_2S_2O_3(aq) + 2HCl(aq) \rightarrow 2NaCl(aq) + SO_2(g) + S(s) + H_2O(l)$$

Analysis

- Gas method: the steeper the initial gradient of the volume–time graph, the faster the rate. Mean rate = volume of gas ÷ time. Smaller chips (larger surface area to volume ratio) give a faster rate, but the same final volume if the same mass is used and the acid is in excess.
- Colour method: rate is proportional to 1/time. Higher temperature or higher concentration gives a shorter time to obscure the cross.
- Explain using collision theory: more frequent collisions (concentration, surface area) and more energetic collisions with energy ≥ activation energy (temperature).

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⚠ **Safety:** Wear eye protection. Sulfur dioxide is toxic and can trigger asthma — work in a well-ventilated room and pour away solutions promptly into a fume-cupboard sink. Do not heat thiosulfate solutions above about 50 °C.

Exam Tip: The disappearing-cross end-point is subjective — use the same person, the same cross and the same viewing height, or use a light sensor and data logger. In the marble method, gas escaping before the bung is inserted is a common error.

CP7 (9.6C) — Identifying the ions in unknown salts using tests for cations and anions

Apparatus: Unknown salt samples (solid and in solution), test tubes and rack, dropping pipettes, nichrome (or platinum) wire loop, concentrated hydrochloric acid (to clean the wire), Bunsen burner, dilute sodium hydroxide solution, dilute hydrochloric acid, dilute nitric acid, barium chloride solution, silver nitrate solution, limewater, damp red litmus paper, water bath.

Tests for cations (positive ions)

1. **Flame test:** clean the wire loop by dipping it in concentrated HCl and holding it in a blue flame until there is no colour. Dip it in the acid then the solid salt and hold it at the edge of a blue flame.
2. **Sodium hydroxide test:** add a few drops of dilute sodium hydroxide solution to a solution of the salt and observe the colour of any precipitate.
3. **Ammonium test:** add sodium hydroxide solution and warm gently; hold damp red litmus paper at the mouth of the tube.

Ion	Flame colour	With NaOH(aq)
Li^+	Red (crimson)	—

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Na^+	Yellow	—
K^+	Lilac	—
Ca^{2+}	Orange-red	White precipitate
Cu^{2+}	Blue-green	Blue precipitate, $Cu(OH)_2$
Fe^{2+}	—	Green precipitate, $Fe(OH)_2$
Fe^{3+}	—	Brown precipitate, $Fe(OH)_3$
Al^{3+}	—	White precipitate that dissolves in excess NaOH
NH_4^+	—	On warming, ammonia gas turns damp red litmus blue

Tests for anions (negative ions)

Ion	Test	Positive result
CO_3^{2-}	Add dilute acid; bubble the gas through limewater	Effervescence; limewater turns milky (CO_2)
SO_4^{2-}	Add dilute HCl, then barium chloride solution	White precipitate of barium sulfate
Cl^-	Add dilute HNO_3, then silver nitrate solution	White precipitate (AgCl)
Br^-	Add dilute HNO_3, then silver nitrate solution	Cream precipitate (AgBr)
I^-	Add dilute HNO_3, then silver nitrate solution	Yellow precipitate (AgI)

(HT) $Cu^{2+}(aq) + 2OH^-(aq) \rightarrow Cu(OH)_2(s)$ $Ba^{2+}(aq) + SO_4^{2-}(aq) \rightarrow BaSO_4(s)$ $Ag^+(aq) + Cl^-(aq) \rightarrow AgCl(s)$

Analysis

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- Test separate portions of the unknown for the cation and the anion, then combine the results to name the salt (e.g. lilac flame + yellow precipitate with acidified $AgNO_3$ = potassium iodide).
- The acid is added first in the anion tests to remove carbonate ions, which would otherwise also give a precipitate. Hydrochloric acid must **not** be used before the silver nitrate test because it adds chloride ions.
- Flame tests can be hard to interpret for mixtures, because one colour (especially yellow from sodium) can mask another. A flame photometer gives a more reliable result and can also measure the concentration of the metal ion.

⚠ **Safety:** Wear eye protection. Concentrated hydrochloric acid is corrosive; sodium hydroxide is an irritant; silver nitrate stains skin and barium chloride is toxic — use small volumes and wash hands. Do not inhale ammonia.

Exam Tip: A dirty wire gives misleading colours — always describe cleaning it. Add NaOH drop by drop so you see the precipitate form before testing whether it dissolves in excess (needed to tell Al^{3+} from Ca^{2+}).

CP8 (9.28C) — Investigating the temperature rise produced by burning alcohols

Apparatus: Spirit burners containing ethanol, propanol, butanol and pentanol (with caps), copper calorimeter (or metal can), clamp stand, thermometer, 100 cm^3 measuring cylinder, balance (0.01 g), heat-proof mat, draught shield, stirrer.

Method

1. Measure 100 cm^3 (100 g) of cold water into the calorimeter and clamp it at a fixed height above the spirit burner. Record the starting temperature of the water.
2. Weigh the spirit burner with its cap on and record the mass.
3. Remove the cap, light the wick and start heating, stirring the water gently. Surround the apparatus with a draught shield.

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4. When the temperature has risen by a set amount (e.g. 20 °C), put the cap on to extinguish the flame. Record the highest temperature reached.
5. Reweigh the burner with its cap on. Mass of alcohol burned = initial mass – final mass.
6. Repeat with each alcohol, using fresh cold water each time, keeping the volume of water, height of calorimeter and wick length the same.

temperature rise per gram = temperature change (°C) ÷ mass of alcohol burned (g)

Analysis

- Compare the temperature rise per gram (or the mass needed for the same temperature rise) for each alcohol.
- Moving along the homologous series (ethanol → propanol → butanol → pentanol), each molecule has an extra CH_2 group, so more energy is released per mole burned and a larger temperature rise is produced per mole.
- Combustion of an alcohol: e.g. $C_2H_5OH + 3O_2 \rightarrow 2CO_2 + 3H_2O$.

⚠ **Safety:** Wear eye protection. Alcohols are highly flammable: keep the bottles capped and away from flames, never refill a burner while lit, and replace the cap to put the flame out. Place the burner on a heat-proof mat.

Exam Tip: The measured temperature rise is lower than expected because energy is lost heating the surroundings and the calorimeter, and incomplete combustion leaves soot on the can. Improve by using a draught shield, a lid, a copper (good conductor) calorimeter and keeping the flame close to the can. Weigh the burner with its cap on, as alcohol evaporates from the wick.

Key Practical Skills

- Identify dependent, independent and control variables.
- Plot graphs with labelled axes, units, and sensible scales; draw lines or curves of best fit and calculate gradients for rates.

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- Estimate uncertainties from the resolution of the balance, burette, thermometer and measuring cylinder.
- Explain random vs systematic errors and how you reduce them (repeats, concordant titres, insulating, using sensors and data loggers).
- Write balanced equations for the reactions in each practical (**(HT)** including ionic equations and half equations).
- Name the equipment precisely — e.g. "volumetric pipette", "Liebig condenser", "gas syringe", "nichrome wire loop".

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